Cognitive Deficit of Deep Learning in Numerosity

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Can DL learn to count?

Subitizing, the ability of babies and primates to glance a small set of objects and know how many without counting.

- A Turing test of Primitivity
- Subject DL to cognitive psychology type experiments to find out
- DL performs poorly on subitizing that kids can do with speed, accuracy and assurance

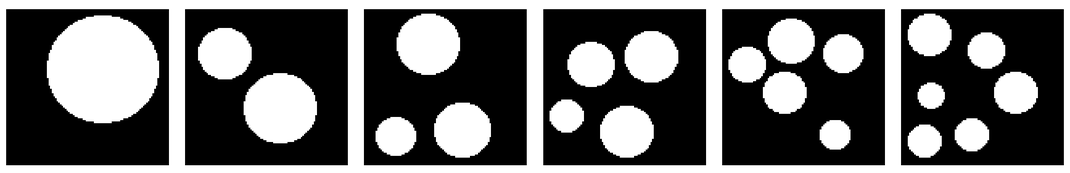
Deep Learning Lacks Abstraction Power

- Long on statistical inference, short on conceptualization.
- Can count specific objects in particular environment
- Cannot see through immaterial variations

Visual Numerosity of DCNNs

Design of Cognitive Experiments

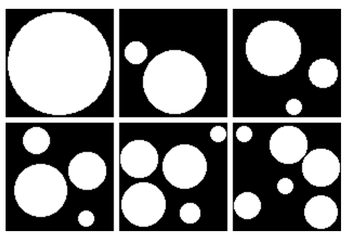
- Raw perception not deliberate counting
- Modeled as a classification instead of regression problem
- Training images are varied visual representations of small numbers



- Achieve a prefect 100% classification rate under the i.i.d. condition

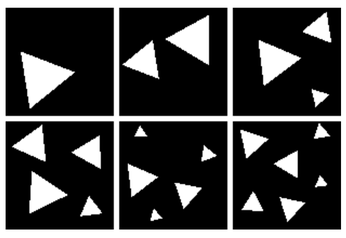
Can it generalize beyond the i.i.d. limitations?

Abstraction in object size (Not quite)



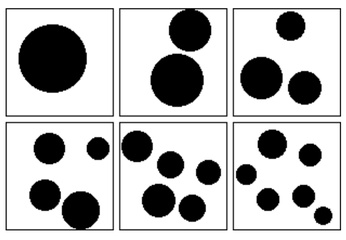
	1	2	3	4	5	6
1	1.00	0	0	0	0	0
2	0	1.00	0	0	0	0
3	0	0.01	0.99	0	0	0
4	0	0	0.04	0.96	0	0
5	0	0	0	0.33	0.67	0
6	0	0	0	0	0.45	0.55

Abstraction in object shape (Fail)



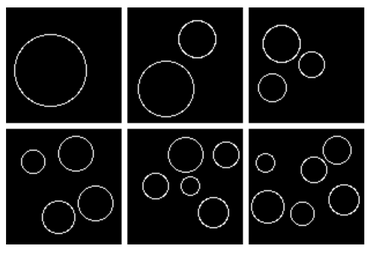
	1	2	3	4	5	6
1	0.33	0.67	0	0	0	0
2	0.03	0.44	0.53	0	0	0
3	0	0.03	0.36	0.61	0	0
4	0	0	0.02	0.29	0.63	0.04
5	0	0	0	0.05	0.36	0.59
6	0	0	0	0	0.07	0.93

Abstraction in object color (Fail)



	1	2	3	4	5	6
1	0.16	0.80	0.04	0	0	0
2	0.01	0.34	0.65	0	0	0
3	0	0.04	0.68	0.28	0	0
4	0	0.01	0.16	0.67	0.16	0
5	0	0	0.06	0.50	0.42	0.02
6	0	0	0	0.18	0.57	0.25

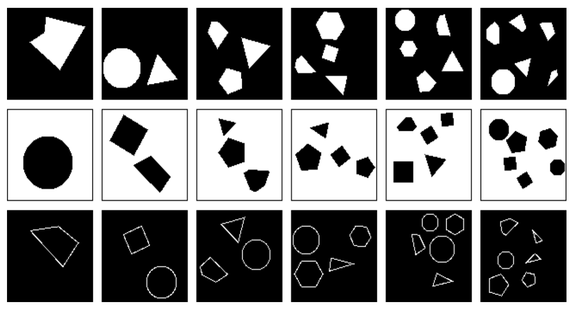
Abstraction in region-boundary duality (Fail)



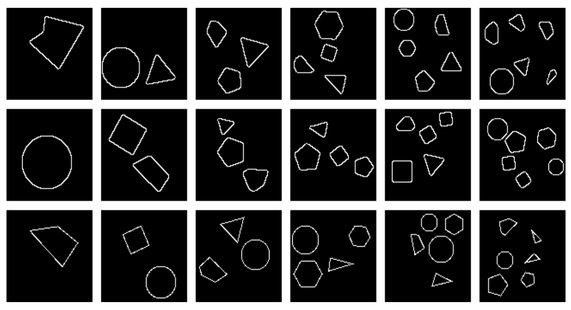
	1	2	3	4	5	6
1	0.01	0.64	0.35	0	0	0
2	0	0	0.42	0.58	0	0
3	0	0	0.01	0.46	0.53	0
4	0	0	0	0.10	0.58	0.32
5	0	0	0	0	0.20	0.80
6	0	0	0	0	0.01	0.99

Generalize with Boundary Representation

Help DCNN with abstraction using the boundary representation of objects instead of the set of pixels.

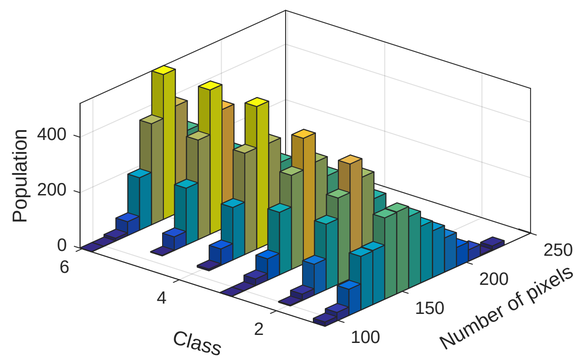


Enriched training images



Unified boundary representations

- n: class label
- c: number of boundary pixels
- c and n are statistically independent.



DCNN still fails to generalize in object size

objects scaled up

	1	2	3	4	5	6
1	0.99	0.01	0	0	0	0
2	0.02	0.98	0	0	0	0
3	0	0.50	0.50	0	0	0
4	0	0	0.79	0.21	0	0
5	0	0	0	0.97	0.03	0
6	0	0	0	0.11	0.86	0.03

objects scaled down

	1	2	3	4	5	6
1	0.69	0.31	0	0	0	0
2	0.02	0.40	0.58	0	0	0
3	0	0.01	0.02	0.90	0.07	0
4	0	0	0	0.01	0.50	0.49
5	0	0	0	0	0.04	0.96
6	0	0	0	0	0.01	0.99

Failure of generalization explained

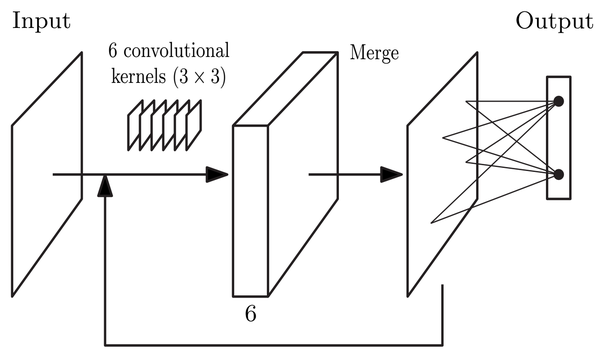
- DCNN Correlates class label n to the average number of boundary pixels per object $\bar{c}$
- Normalizing c against n creates a correlation between n and $\bar{c}$
- Normalizing $c\bar{c}$ against n creates a correlation between n and c
- Dilemma: impossible to make both c and $\bar{c}$ independent of n.

Deterministic General DCNN Solution

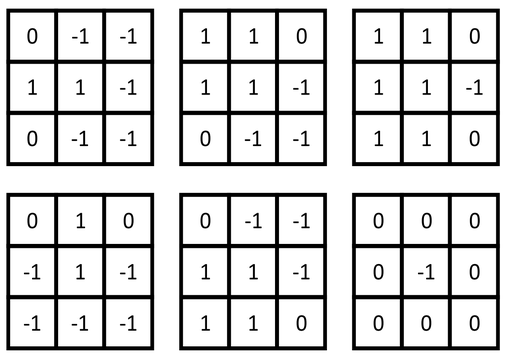
Cognitive deficit of DL is with the back propagation methodology, not the computational machinery of deep convolutional neural networks.

Design a RCNN that can subitize

- Build into CNN kernels a mechanism of mathematical morphology
- Recursively removing boundary pixels of the objects in parallel
- Reduce (abstract) objects of complex shapes to single pixels
- Although the geometry of the image signal keeps simplifying, the topology of the objects is preserved, thus maintaining the object count an invariant.



RCNN architecture



RCNN kernels

Is the hand-crafted RCNN trainable? No!

Can Learn with preexisted cognitive implements

- Reduce the cognitive task of subitizing to identifying connected components; the latter is the essential feature of the former.
- Decomposing subitizing into three subfunctions: 1).one layer erosion, 2).connected component; 3).counting.